

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech (CE) Examination****May 2017****CE219 Object Oriented Programming using Java****Date: 31.05.2017, Wednesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections and both must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Write entire option in case of multiple choice and matching questions.

SECTION – I**Q – 1 (A) Fill in the blanks.****[06]**

1. Vertical division of system is called _____.
2. Full form of UML is _____.
3. In system description nouns are identified as _____ in class diagram.
4. Systems can interact with each other using _____.
5. Smallest subsystem which cannot be further divided is called _____.
6. State diagram is fall in the group of _____ diagrams.

(B) List four concepts of OO.**[02]****(C)** Explain join and fork with respect to activity diagram.**[03]****Q - 2 Answer the following questions. (Any three)****[12]**

- (A) Explain qualified association and association class.
- (B) What is subsystem? How to identify the subsystem? How subsystems are interacting with each other? List out the relationships between subsystems.
- (C) List purpose of modeling.
- (D) Name two kind of control flow in software system. Explain them in detail.

Q – 3 Answer the following questions.**(A)** Draw use case diagram for library management system.**[07]****(B)** What do you mean by Layer? Explain with example. Mention two forms of layered architecture.**[05]****OR****Q – 3 Answer the following questions.****(A)** List and explain type of relationships in class diagram.**[04]****(B)** List phases of SDLC.**[04]****(C)** Draw sequence diagram of product buying.**[04]**

SECTION – II**Q – 4 Answer the following short questions.**

- (A) Define following [02]
 1. Bytecode
 2. Applet
- (B) “Java is platform independent”. Given statement is true or false? Justify your answer. [02]
- (C) Differentiate array list and vector. [02]
- (D) Pen down two ways to create a thread. [02]
- (E) What is stream? Write one similarity and one difference between character stream and byte stream? [03]

Q - 5 Answer the following questions. (Any three) [12]

- (A) Differentiate method overloading and overriding.
- (B) Draw and explain Applet life cycle.
- (C) Explain multiprogramming and multitasking.
- (D) Write a Java program to print following pattern. Take number of rows as input from user.
- ```

1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1

```

**Q - 6 Answer the following questions.**

- (A) Explain Single level and multilevel inheritance with example. [05]
- (B) Which is the super class of all type exceptions in Java? What happen when exceptions are not handled? Explain how to handle exception using try-catch, multiple catch and nested try in Java. [05]
- (C) What will be the output of following code? [02]

```

class StrEqual {
 public static void main(String []args) {
 String s1 = "hi";
 String s2 = new String("hi");
 String s3 = "hi";
 if(s1 == s2) {
 System.out.println("s1 and s2 equal");
 } else {
 System.out.println("s1 and s2 not equal");
 }
 if(s1 == s3) {
 System.out.println("s1 and s3 equal");
 } else {
 System.out.println("s1 and s3 not equal");
 }
 }
}

```

```
 }
 }
}
```

**OR**

**Q - 6 Answer the following questions.**

- (A) Write a Java program to sort Array list of numbers. [05]
- (B) Write a Java program that illustrate the interface inheritance. Interface P is extended by P1 and P2 interfaces. Interface P12 extends both P1 and P2. Each interface declares one method and one constant. Create one class that implements P12. By using the object of the class invokes each of its method and displays constant. [05]
- (C) What four access modes can you specify for a random-access file? [02]

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